

FIGURE P2-16

P2-17 $i_2 = 10 \text{ mA}$, $i_3 = -15 \text{ mA}$, and i_1 and i_5 .

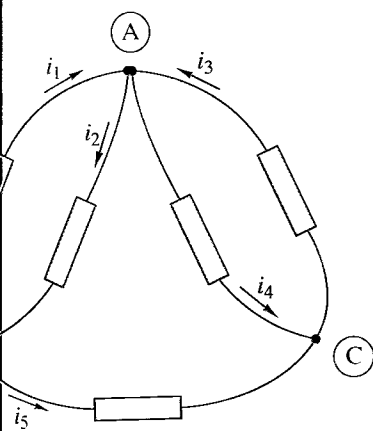


FIGURE P2-17

Use the passive sign convention to assign voltage and current to the elements in Figure P2-17. Write the connection equations using these voltage and current variables.

What can be said about the voltages across the elements?

Write the connection equations for a three-node circuit are:

$$\begin{aligned} -i_1 + i_2 - i_4 &= 0 \\ -i_2 - i_3 + i_5 &= 0 \\ i_1 + i_3 + i_4 - i_5 &= 0 \end{aligned}$$

Use the circuit diagram and indicate the reference direction for the element currents.

3 COMBINED CONSTRAINTS

consisting of independent sources and linear elements. Use the element constraints and connection constraints to find the selected signal variables.

2-7, 2-8, 2-9, 2-10 and Exercise 2-6

Find i_x in Figure P2-20.

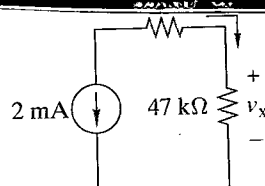


FIGURE P2-20

2-21 Find v_x and i_x in Figure P2-21.

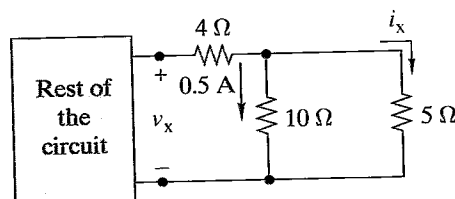


FIGURE P2-21

2-22 In Figure P2-22:

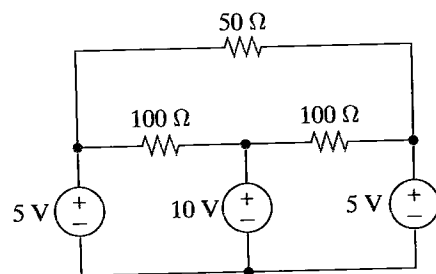


FIGURE P2-22

- Assign a voltage and current variable to every element.
- Use KVL to find the voltage across each resistor.
- Use Ohm's law to find the current through each resistor.
- Use KCL to find the current through each voltage source.

2-23 Find the power dissipated in the 1.5-kΩ resistor in Figure P2-23.

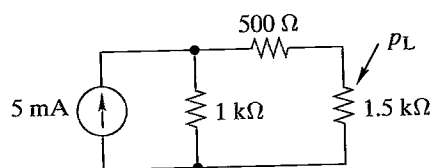


FIGURE P2-23

2-24 Figure P2-24 shows a subcircuit connected to the rest of the circuit at four points.

- Use element and connection constraints to find v_x and i_x .
- Show that the sum of the currents into the rest of the circuit is zero.

- (c) Find the voltage v_A with respect to the ground in the circuit.

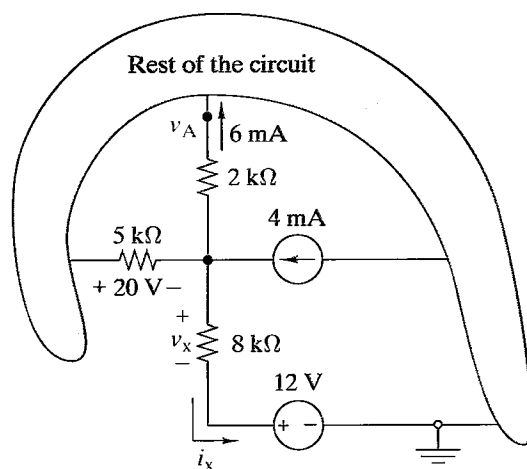


FIGURE P2-24

- 2-25 In Figure P2-25 $i_x = -0.5$ mA. Find the value of R .

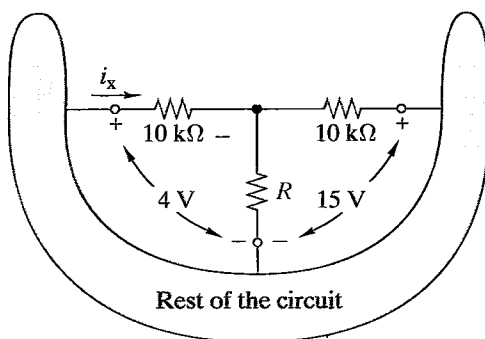


FIGURE P2-25

- 2-26 Figure P2-26 shows a resistor with one terminal connected to ground and the other connected to an arrow. The arrow symbol is used to indicate a connection to one terminal of a voltage source whose other terminal is connected to ground. The label next to the arrow indicates the source voltage at the ungrounded terminal. Find the voltage across, current through, and power dissipated in the resistor.

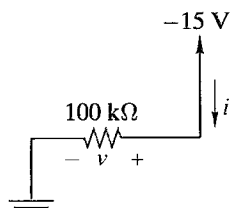


FIGURE P2-26

OBJECTIVE 2-4 EQUIVALENT CIRCUITS (SECT. 2-4)

- (a) Given a circuit consisting of linear resistors, find equivalent resistance between a specified pair of terminals.
 (b) Given a circuit consisting of a source-resistor combination, find an equivalent source-resistor circuit.
 See Examples 2-11, 2-12 and Exercises 2-7, 2-8, 2-9

- 2-27 Find the equivalent resistance R_{EQ} in Figure P2-27.

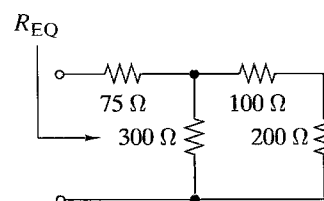


FIGURE P2-27

- 2-28 Find the equivalent resistance R_{EQ} in Figure P2-28.

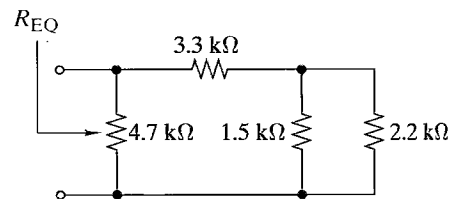


FIGURE P2-28

- 2-29 Find R_{EQ} in Figure P2-29 when the switch is open. Repeat when the switch is closed.

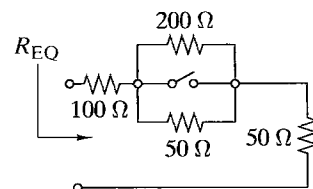


FIGURE P2-29

- 2-30 Show how the circuit in Figure P2-30 could be connected to achieve a resistance of 100 Ω, 200 Ω, 150 Ω, 50 Ω, 25 Ω, 33.3 Ω, and 133.3 Ω.

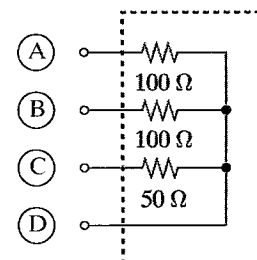


FIGURE P2-30

P2-31 find the equivalent resistance between terminals A, B, A-C, A-D, B-C, B-D, and C-D.

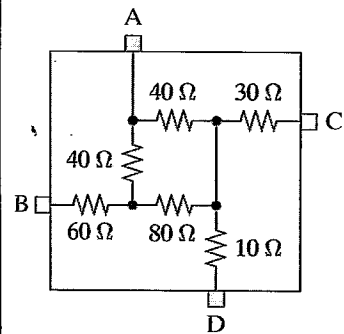


FIGURE P2-31

value of R_L in Figure P2-32 so that $R_{EQ} = 25 \text{ k}\Omega$. $R_{EQ} = 20 \text{ k}\Omega$.

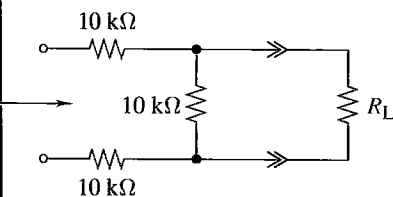


FIGURE P2-32

more than four 1-k Ω resistors, show how the equivalent resistors can be constructed: 1.5 k Ω , 333 Ω , 250 Ω , and 400 Ω .

equivalent practical voltage source at terminals A and B in Figure P2-34.

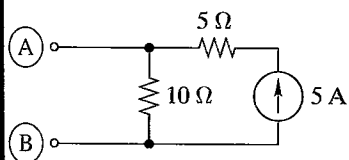


FIGURE P2-34

P2-35 the $i-v$ characteristic of network N is $v = 10i$. Find the equivalent practical current source at terminals A and B in Figure P2-35.

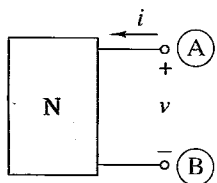


FIGURE P2-35

2-36 Select the value of R_x in Figure P2-36 so that $R_{EQ} = 60 \text{ k}\Omega$.

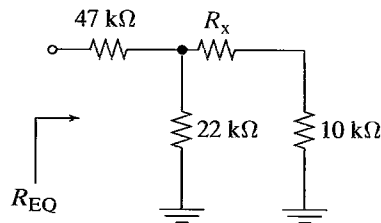


FIGURE P2-36

2-37 Two 10-k Ω potentiometers (a potentiometer is a variable resistor whose value between the two ends is 10 k Ω and between one end and the wiper—the third terminal—can range from 0 Ω to 10 k Ω) are connected as shown in Figure P2-37. What is the range of R_{EQ} ?

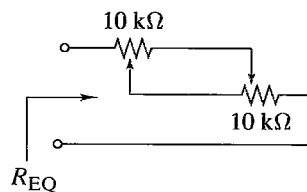


FIGURE P2-37

2-38 Select the value of R in Figure P2-38 so that $R_{AB} = R_L$.

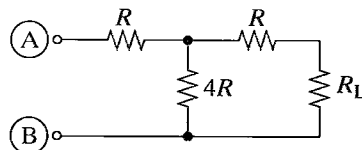


FIGURE P2-38

2-39 What is the range of R_{EQ} in Figure P2-39?

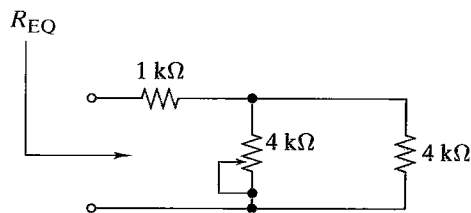


FIGURE P2-39

2-40 Find the equivalent resistance between terminals A and B in Figure P2-40.

- 2-57 Use source transformations in Figure P2-57 to relate v_O to v_1 , v_2 , and v_3 .

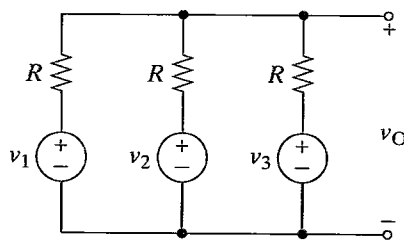


FIGURE P2-57

- 2-58 The current through R_L in Figure P2-58 is 40 mA. Use source transformations to find R_L .

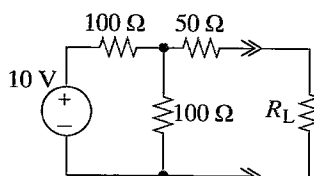


FIGURE P2-58

- 2-59 Select R_x so that 25 V are across it in Figure P2-59.

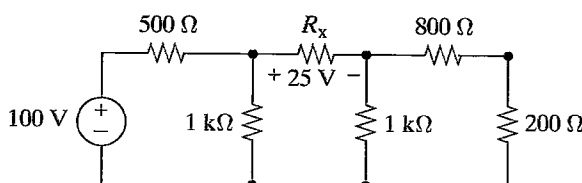


FIGURE P2-59

- 2-60 The box in the circuit in Figure P2-60 is a resistor whose value can be anywhere between 8 kΩ and 80 kΩ. Use circuit reduction to find the range of values of v_x .

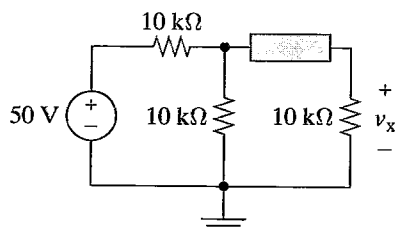


FIGURE P2-60

OBJECTIVE 2-7 COMPUTER-AIDED CIRCUIT ANALYSIS (SECT. 2-7)

Given an appropriate linear circuit, use circuit simulation and/or computational software to solve for the desired response. See Examples 2-24, 2-25 and Appendix B.

- 2-61 A circuit is found to have the following element connection equations:

$$\begin{aligned} v_1 &= 24 \text{ V} & -v_1 + v_2 + v_3 &= 0 \\ v_2 &= 8k i_2 & -v_3 + v_4 + v_5 &= 0 \\ v_3 &= 5k i_3 & i_1 + i_2 &= 0 \\ v_4 &= 4k i_4 & -i_2 + i_3 + i_4 &= 0 \\ v_5 &= 16k i_5 & -i_4 + i_5 &= 0 \end{aligned}$$

Use MATLAB to solve for all of the unknown voltages and currents associated with this circuit. Sketch one possible circuit that matches the given equations.

- 2-62 Consider the circuit of Figure P2-62. Use MATLAB to find all of the voltages and currents in the circuit.

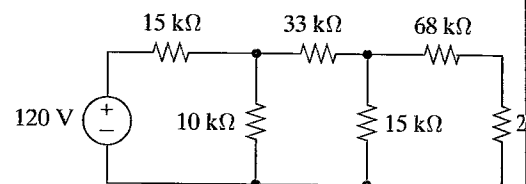


FIGURE P2-62

- 2-63 Consider the circuit of Figure P2-62 again. Use OrCAD to find all of the voltages and currents in the circuit.
- 2-64 Use OrCAD to find all the power dissipated or produced in the circuit of Figure P2-62. Verify that the total power in the circuit is zero.

INTEGRATING PROBLEMS

2-65 Nonlinear Device Characteristics

The circuit in Figure P2-65 is a parallel combination of a linear resistor and a varistor whose i - v characteristic is $i_v = 2.6 \times 10^{-5} v^3$. For a small voltage, the varistor current is quite small compared to the resistor current. For large voltages, the varistor dominates because its current increases more rapidly with voltage.

- Plot the i - v characteristic of the parallel combination.
- State whether the parallel combination is linear or nonlinear, active or passive, and bilateral or nonbilateral.
- Find the range of voltages over which the resistor current is at least 10 times as large as the varistor current.
- Find the range of voltages over which the varistor current is at least 10 times as large as the resistor current.

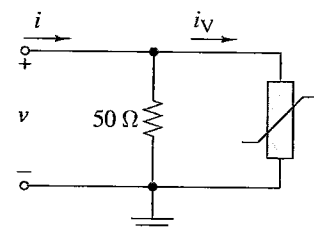


FIGURE P2-65

- 3-9 (a) Formulate mesh-current equations for the circuit in Figure P3-9.

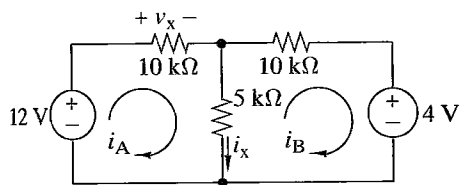


FIGURE P3-9

- (b) Solve for i_A and i_B .
(c) Use these results to find v_x and i_x .

- 3-10 (a) Formulate mesh-current equations for the circuit in Figure P3-10.

- (b) Solve for i_A and i_B .
(c) Use these results to find v_x and i_x .

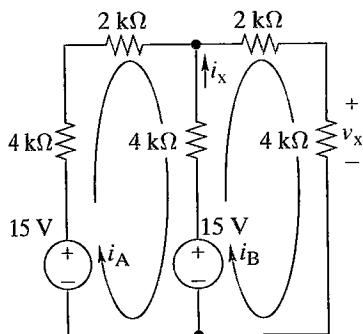


FIGURE P3-10

- 3-11 (a) Formulate mesh-current equations for the circuit in Figure P3-11.

- (b) Formulate node-voltage equations for the circuit in Figure P3-11.

- (c) Which set of equations would be easier to solve? Why?
(d) Using MATLAB, find v_x and i_x in terms of the mesh-current variables.

- (e) Using MATLAB, find v_x and i_x in terms of the node-voltage variables.

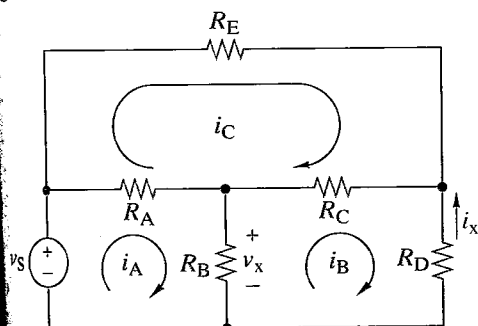


FIGURE P3-11

- 3-12 (a) Formulate mesh-current equations for the circuit in Figure P3-12.

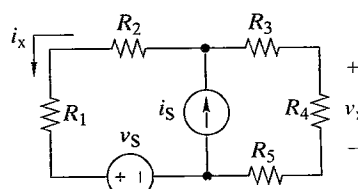


FIGURE P3-12

- (b) Solve for v_x and i_x when $R_1 = 200 \Omega$, $R_2 = 300 \Omega$, $R_3 = 50 \Omega$, $R_4 = 250 \Omega$, $R_5 = 200 \Omega$, $i_S = 100 \text{ mA}$, and $v_S = 25 \text{ V}$.
(c) Find the total power dissipated in the circuit.

- 3-13 (a) Formulate mesh-current equations for the circuit in Figure P3-13.

- (b) Use MATLAB to find symbolic expressions for v_x and i_x in terms of the parameters in the circuit.

- (c) Find numeric values for v_x and i_x when $R_1 = R_2 = 4.7 \text{ k}\Omega$, $R_3 = 2.2 \text{ k}\Omega$, $R_4 = 1 \text{ k}\Omega$, $i_S = 3 \text{ mA}$, $v_{S1} = 10 \text{ V}$, and $v_{S2} = 2 \text{ V}$.
(d) Find the power supplied by v_{S1} .

- (e) Use OrCAD to verify that your solutions to parts (c) and (d) are correct.

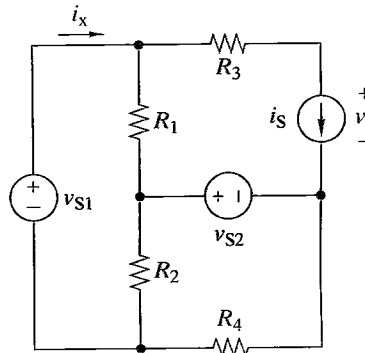


FIGURE P3-13

- 3-14 The circuit in Figure P3-14 seems to require two supermeshes since both current sources appear in two meshes. However, sometimes rearranging the circuit diagram will eliminate the need for a supermesh.

- (a) Show that supermeshes can be avoided in Figure P3-14 by rearranging the connection of resistor R_6 .

- (b) Formulate mesh-current equations for the modified circuit as redrawn in part (a).

- (c) Solve for v_x when $R_1 = R_2 = R_3 = R_4 = 2 \text{ k}\Omega$, $R_5 = R_6 = 1 \text{ k}\Omega$, $i_{S1} = 40 \text{ mA}$, and $i_{S2} = 20 \text{ mA}$.

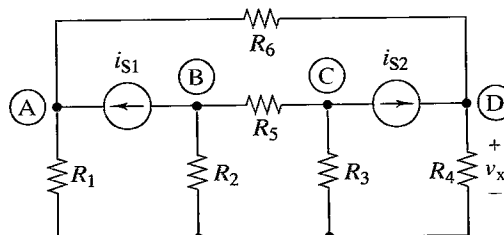


FIGURE P3-14

- 3-15(a) Formulate mesh-current equations for the circuit in Figure P3-15.
 (b) Formulate node-voltage equations for the circuit in Figure P3-15.
 (c) Which set of equations would be easier to solve? Why?
 (d) Find v_x and i_x using whichever method you prefer.

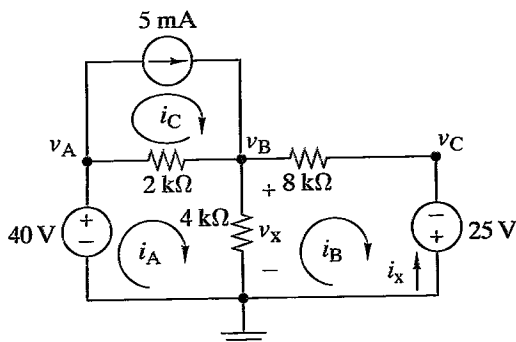


FIGURE P3-15

- 3-16 Use mesh-current equations, node-voltage equations, or simple engineering intuition to find the input resistance of the circuit in Figure P3-16.

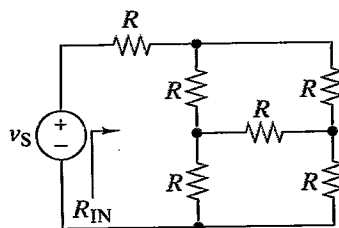


FIGURE P3-16

- 3-17 In Figure P3-17 all of the resistors are $1\text{ k}\Omega$ and $v_S = 10\text{ V}$. The voltage at node C is found to be $v_C = -2\text{ V}$ when node B is connected to ground. Find the node voltages v_A and v_D , and the mesh currents i_A and i_B .

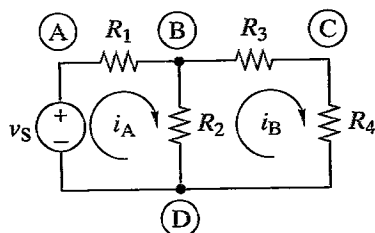


FIGURE P3-17

- 3-18 Use Figure P3-17 and MATLAB to solve the following problems:
 (a) Using mesh-current analysis, find a symbolic expression for i_A in terms of the circuit parameters.
 (b) Compute the ratio v_S/i_A .

- (c) Find a symbolic expression for the equivalent resistance of the circuit by combining resistors in series and parallel. Compare your answer to the results from part (b).

- 3-19 Use OrCAD to find the node voltages v_A and v_B and mesh currents i_A , i_B , and i_C in Figure P3-19.

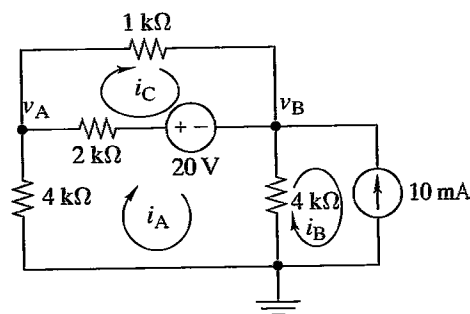


FIGURE P3-19

- 3-20(a) Use MATLAB and mesh-current analysis to solve for the mesh currents i_A , i_B , i_C , and i_D in Figure P3-20.
 (b) Solve for the same currents using node-voltage analysis and compare the effort required with each technique.
 (c) Use OrCAD to verify your results in the previous two parts.

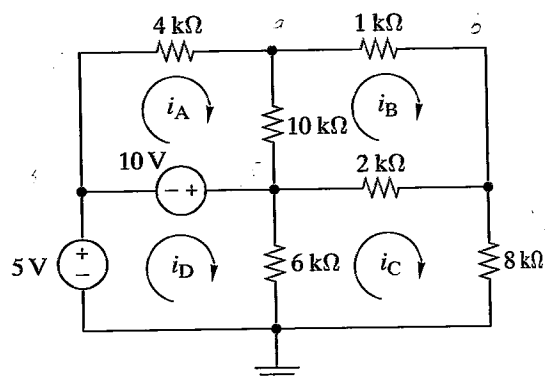


FIGURE P3-20

OBJECTIVE 3-2 LINEARITY PROPERTIES (SECT. 3-3)

- (a) Given a circuit containing linear resistors and one independent source, use the proportionality principle to find selected signal variables.
 (b) Given a circuit containing linear resistors and two or more independent sources, use the superposition principle to find selected signal variables.

See Examples 3-10, 3-11, 3-12 and Exercises 3-18, 3-19, 3-20, 3-21, 3-22

divider and calculates the power in the load using $I_L^2 R_L$. He figures correctly what the parallel voltage would be across the Norton circuit and the load v_L , and then calculates $P_S = I_N \times v_L$. What grade would you give him?

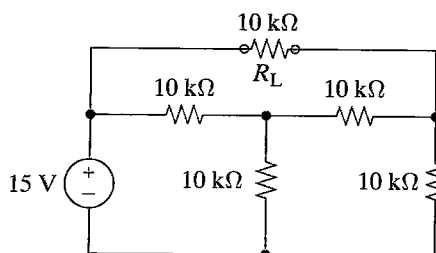


FIGURE P3-44

3-45 The Thévenin equivalent parameters of a practical voltage source are $v_T = 30$ V and $R_T = 300 \Omega$. Find the smallest load resistance for which the load voltage exceeds 15 V.

3-46 Use a sequence of source transformations to find the Thévenin equivalent at terminals A and B in Figure P3-46.

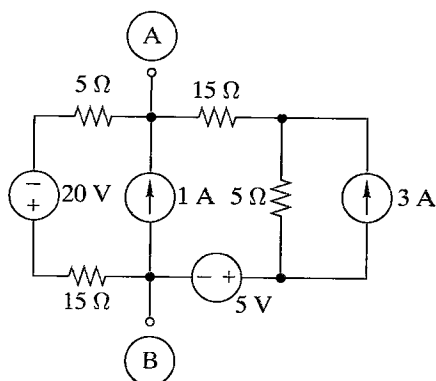


FIGURE P3-46

3-47 The circuit in Figure P3-47 provides power to a number of loads connected in parallel. The circuit is protected by a $\frac{3}{4}$ -mA fuse with a nominal $100\text{-}\Omega$ resistance. Each load is $10\text{ k}\Omega$. What is the maximum number of loads the circuit can drive without blowing the fuse?

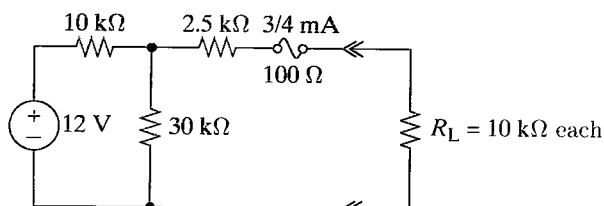


FIGURE P3-47

3-48 Find the Thévenin equivalent at terminals A and B in Figure P3-48.

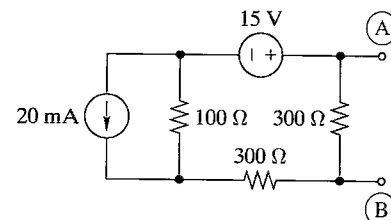


FIGURE P3-48

3-49 A nonlinear resistor is connected across a two-terminal source whose Thévenin equivalent is $v_T = 10$ V and $R_T = 500 \Omega$. The i - v characteristic of the resistor is $i = 10^{-4}(v + 2v^{3.3})$. Use the MATLAB function `fsolve` to find the operating point for this circuit and determine the voltage across, the current through, and the power dissipated in the nonlinear resistor.

3-50 A blue light-emitting diode (LED) is connected across a two-terminal source whose Thévenin equivalent is $v_T = 10$ V and $R_T = 20 \Omega$. The i - v characteristic of the LED is $i = 10^{-12}(e^{10v} - 1)$. Figure P3-50 shows the LED's characteristic. Using either MATLAB or a graphical method, determine the voltage across and current through the LED.

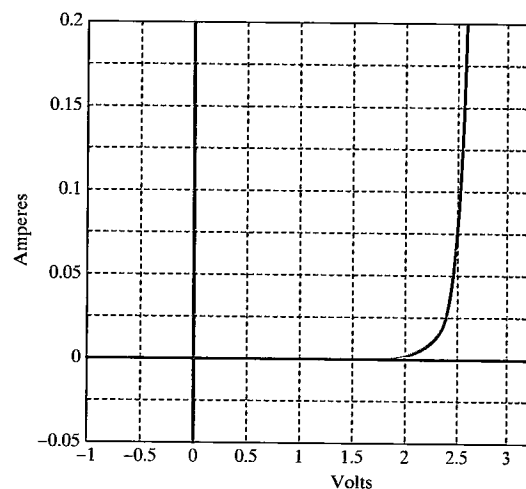


FIGURE P3-50

3-51 Find the Norton equivalent seen by R_L in Figure P3-51. Select the value of R_L so that 100 mW is delivered to the load.

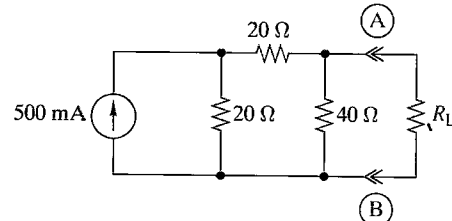


FIGURE P3-51

-  **4-40** Find the output v_O in terms of the input v_i in Figure P4-40. Verify your answer using OrCAD.

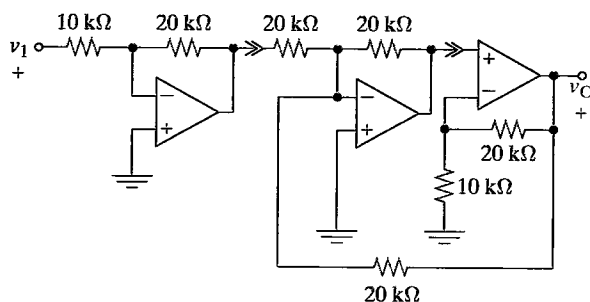


FIGURE P4-40

OBJECTIVE 4-3 OP AMP CIRCUIT DESIGN AND EVALUATION (SECT. 4-5)

Given an input-output relationship, use resistors and OP AMPs to design one or more circuits that implement the relationship within the stated constraints. Choose the better of two working circuits to meet certain constraints.

See Design Examples 4-12, 4-15, 4-20, 4-21, 4-22, 4-23, 4-24, Design Exercises 4-12, 4-16, 4-22, 4-28, 4-29, 4-30, Evaluation Examples 4-19, 4-20, 4-21, 4-22

- 4-41**  Design an OP AMP amplifier with a voltage gain of -10 and an input resistance greater than $10\text{ k}\Omega$ using standard 5% resistance values less than $300\text{ k}\Omega$.
- 4-42**  Design an OP AMP amplifier with a voltage gain of 4 using only $10\text{-k}\Omega$ resistors and one OP AMP.
- 4-43**  Using a single OP AMP, design a circuit with inputs v_1 and v_2 and an output $v_O = 2v_2 - 3v_1$. The input resistance seen by each input should be greater than $5\text{ k}\Omega$.
- 4-44**  Design a differential amplifier with inputs v_1 and v_2 and an output $v_O = 10(v_2 - v_1)$ using only one OP AMP. All resistances must be between $5\text{ k}\Omega$ and $200\text{ k}\Omega$.
- 4-45**  Using no more than two OP AMPs, design an OP AMP circuit with inputs v_1 , v_2 , and v_3 and an output $v_O = -3v_1 + 2v_2 - 5v_3$.
-  **4-46**  Design the interface circuit in Figure P4-46 so that 10 mW is delivered to the $100\text{-}\Omega$ load. Repeat for a $100\text{-k}\Omega$ load. Verify your designs using OrCAD.

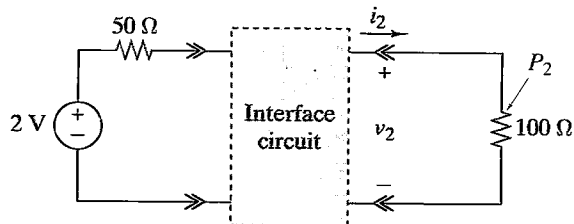


FIGURE P4-46

- 4-47 **D** Design the interface circuit in Figure P4-47 so the output is $v_2 = 150v_1 + 1.5$ V.

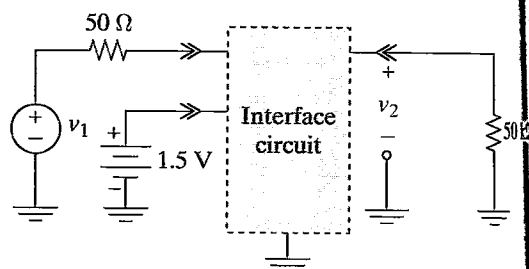
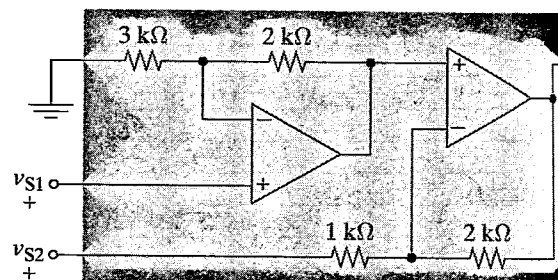


FIGURE P4-47

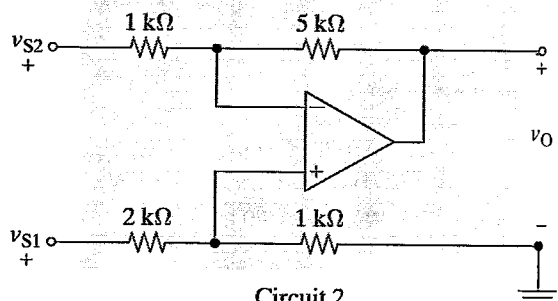
- 4-48 D** (a) Design a circuit that can produce $v_o = 1000$ V using two OP AMPs.
(b) Repeat using only one OP AMP.
- 4-49 E** A requirement exists for an OP AMP circuit with input-output relationship

$$v_O = 5v_{S1} - 2v_{S2}$$

Two proposed designs are shown in Figure P4-49. As project engineer, you must recommend one of these circuits for production. Which of these circuits would you recommend for production and why? (*Hint: First verify that the circuits perform the required function.*)



Circuit 1



Circuit 2

FIGURE P4-49

- 4-50**  A requirement exists for an OP AMP circuit that will deliver 12 V to a 1-k Ω load using a 4-V source as an input.

- 4-56 **D** A small pressure transducer has the characteristics shown in Figure P4-56. Design an interface that will operate between 7 and 32 psi. An input of 7 psi should produce -5 V and 32 psi should produce 5 V. The transducer is modeled as a voltage source in series with a $500\ \Omega$ resistor that can vary by $\pm 75\ \Omega$ depending on the pressure. The OP AMPs you must use have a maximum closed-loop gain of 2000. Your available bias source is 5 V.

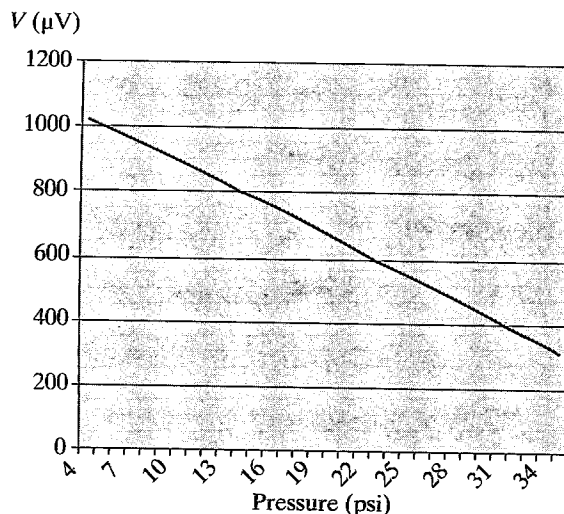
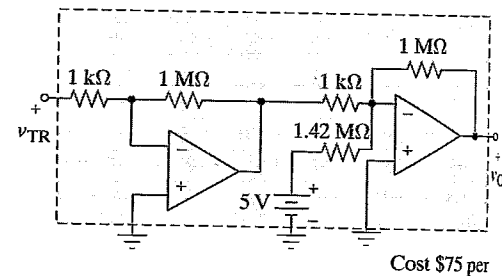


FIGURE P4-56

- 4-57 **D** A medical-grade pressure transducer has been developed for use in invasive blood pressure monitoring. The output voltage of the transducer is $v_{TR} = (0.05P - 0.75)$ mV, where P is pressure in mm Hg. The output resistance of the transducer is $300\ \Omega$. The blood pressure measurement is to be an input to an existing multisensor monitoring system. This system treats a 1 V input as a blood pressure of 20 mm Hg and a 10 V input as 200 mm Hg. Design an OP AMP circuit to interface the new pressure transducer with the existing monitoring system.
- 4-58 **D** The acid/alkaline balance of a fluid is measured by the pH scale. The scale runs from 0 (extremely acid) to 14 (extremely alkaline), with $\text{pH } 7$ being neutral. A pH electrode is a sensor that produces a small voltage that is directly proportional to the pH of the fluid in a test chamber. For a certain pH electrode, the proportionality factor is 60 mV/pH. A preamplifier is needed to interface this sensor with a variety of laboratory instruments. The output of the preamp must be 1 V when the sensor is immersed in a test solution with $\text{pH} = 4$ and 1.75 V when it is immersed in a solution with $\text{pH} = 7$. Design an amplifier to meet these requirements.
- 4-59 **D** A photoresistor varies from $100\ \Omega$ in bright sunlight to $200\ \text{k}\Omega$ in total darkness. Design a suitable circuit using the photoresistor so that total darkness produces 5 V while bright sunlight produces -5 V, regardless of the load. You have a 5 -V source and a ± 15 -V source to power any OP AMP you may need.

- 4-60 **E** Your engineering firm needs an instrumentation amplifier that provides the following input-output relation $v_O = 10^6 v_{TR} - 3.5$ V. The transducer is modeled as a voltage source in series with a resistor that varies with transducer voltage from $40\ \Omega$ to $750\ \Omega$. A vendor is offering the amplifier shown in Figure P4-60 and agrees to make a single change to the amplifier, if needed, for no cost. Would you recommend buying it? Explain your rationale.



Instrumentation Amplifier Model 54

FIGURE P4-60

- 4-61 The OP AMP in Figure P4-61 operates as a comparator. Find the output voltage when $v_S = 2$ V. Repeat for -2 V and $v_S = 6$ V.

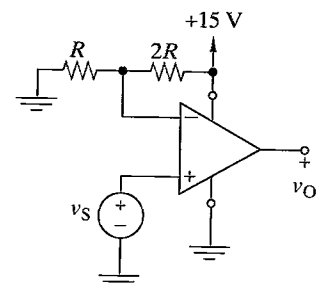


FIGURE P4-61

- 4-62 The circuit in Figure P4-62 has $V_{CC} = 15$ V and $v_N = -2$ V. Sketch the output voltage v_O on the graph for $0 \leq t \leq 2$ s for $v_S(t) = 10 \sin(2\pi t)$ V.

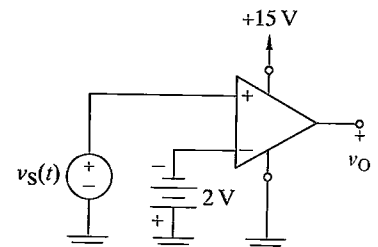


FIGURE P4-62

- 4-63 **D** A signal varies as $v_S(t) = 10(1 - e^{-100t})$ V. Design a comparator circuit that will switch from 0 to 5 V in 2 s.